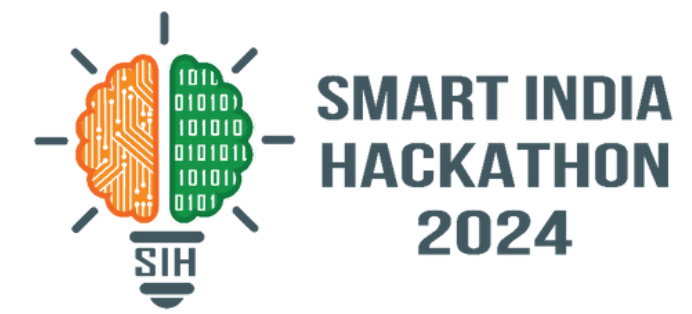


# SMART INDIA HACKATHON 2024



**Problem Statement ID – SIH1645**

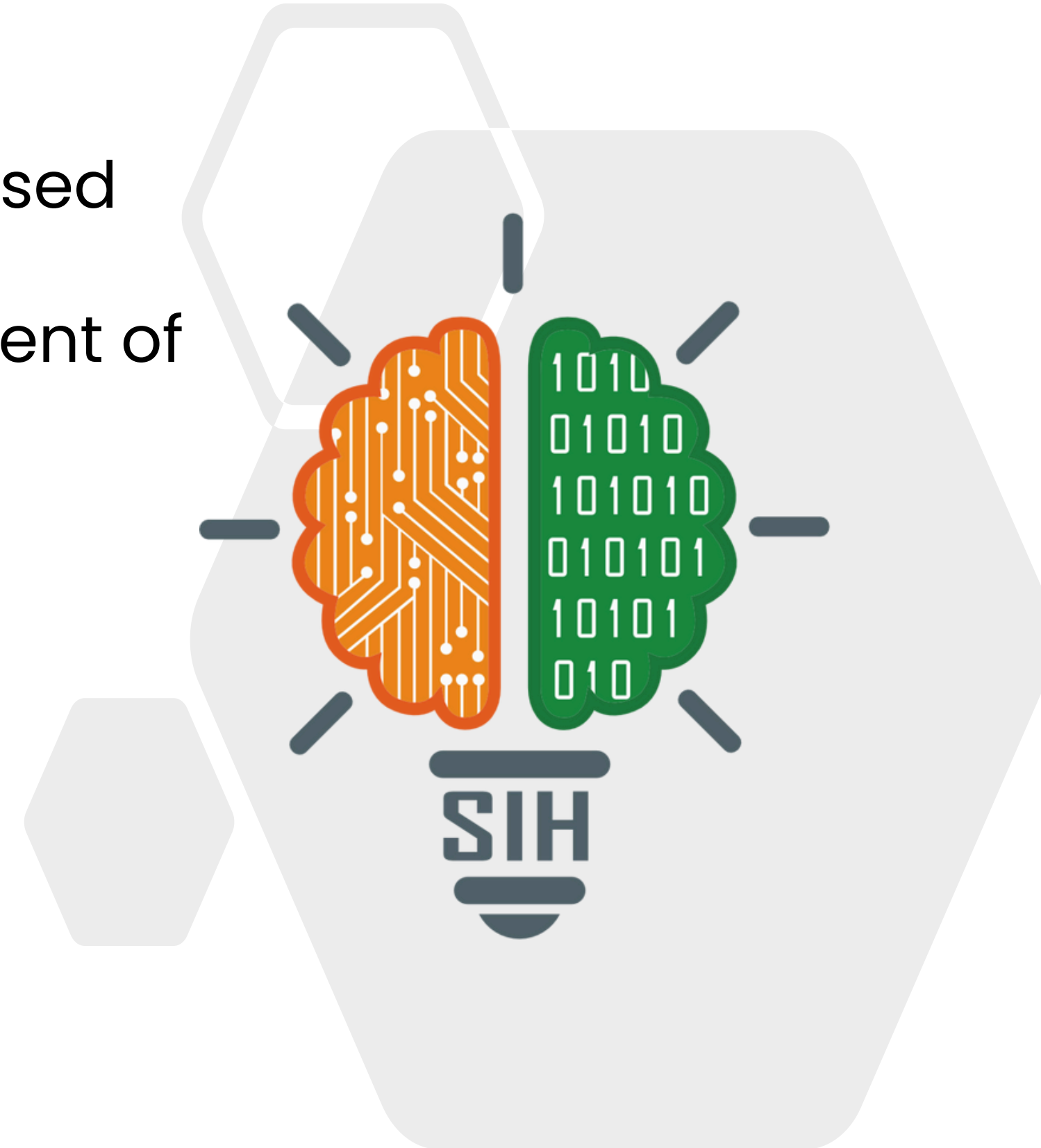
**Problem Statement Title –** An app and web based software for Productivity and safety management of coal mines.

**Theme –** Smart Automation

**PS Category–** Software

**Team ID–** 289

**Team Name(Registered on Portal) –** Arize



**SOLUTION**

**CoalWorks is a mining operations software that combines App, Web, and IoT for a smoother, safer, and more efficient mining process.**

- **Web dashboard for supervisors** to monitor activities, **plan rounds, track tasks, and handover shifts**
- **AI-powered assistant suggests round planning and assigns the right personnel** to handle issues
- **OCR is used to scan and convert paper logbooks into easy template based digital forms**
- **Mobile app for workers** to view **shift details and log events** like issues, alerts, injuries, etc.
- **Admin Dashboard** for easy integration with existing **DGMS-compliant SMPs and logbooks**
- **Easy data migration** from existing **ERP systems using data mapping and APIs**

**We solve the problem with complete mining software tailored for India, offering digital logging, safety management, and AI/IoT for efficiency.**

CoalWorks is **50% completed**; testing and validation are ongoing.

**WHY WE STAND OUT ?**

**Predictive Simulation and Analysis AI Engine**  
Supervisors can simulate mine operations, estimate needs for better decisions.

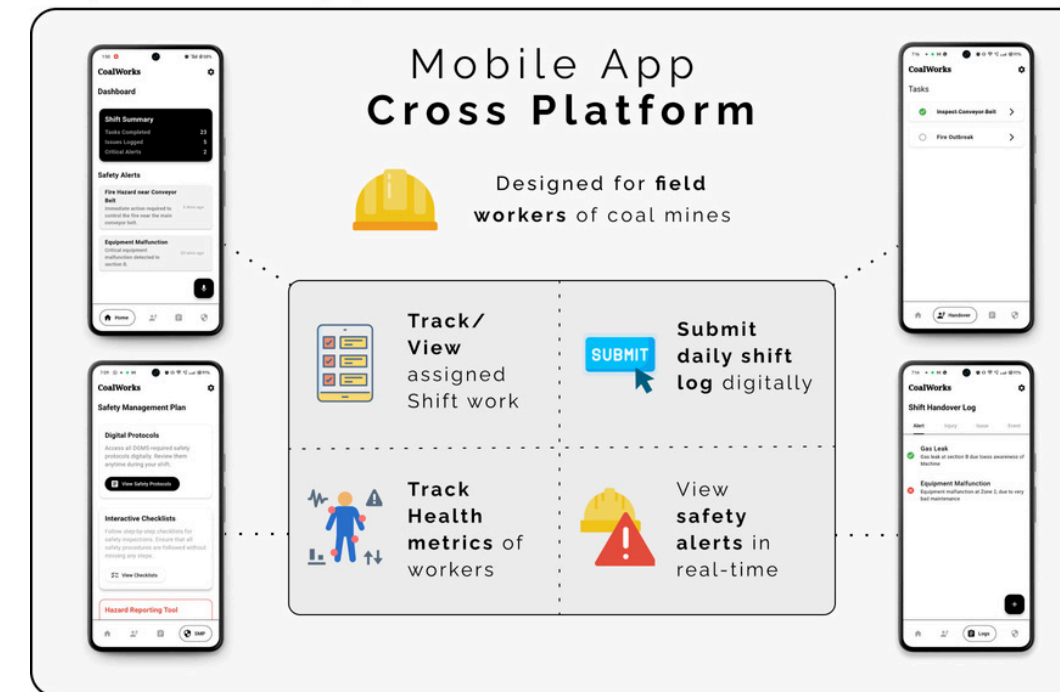
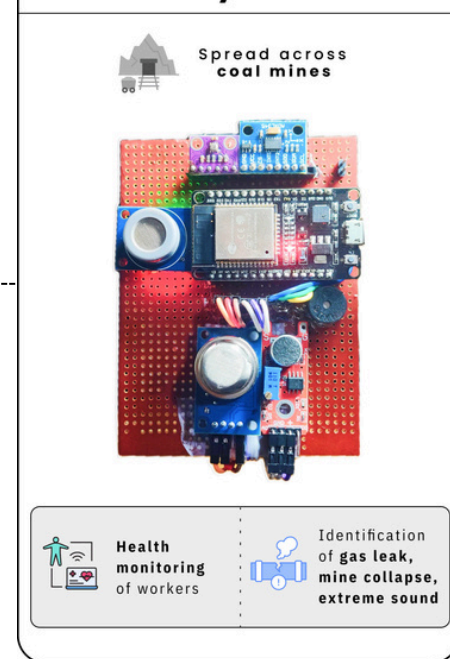
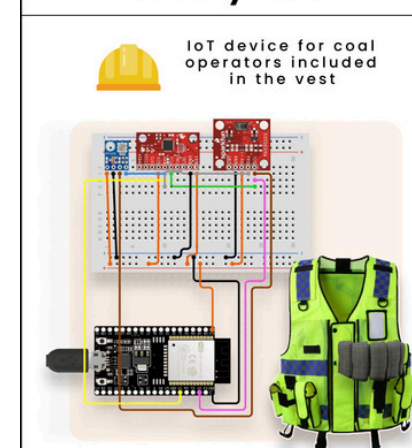
**Optimizing Workflows and Resources**  
Our AI predicts hazards, plans shifts, assigns tasks, optimizes resources and safety.

**Cross-Platform Solution with Docker Integration**  
Our Product ensure consistent performance and accessibility across different environments.

**IoT Scalable IoT Deployment for cross validation and safety**  
Our IoT devices ensure safety and productivity in coal mines, even if no internet.

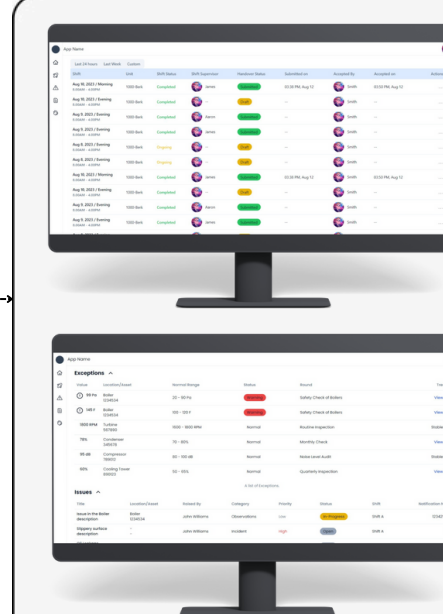
**PROTOTYPE**

We support **10+ languages** with an easy **UI/UX**, hassle free logging of tasks using **smart voice controls**

**Operator App****Safety nodes****Safety Vest**

Health monitoring of workers

Parameters:  
SPO2, Pulse, Fall detection

**Supervisor App****Desktop Application**

Designed for Mine supervisors and Admin

OCR to scan SMP and logs to form

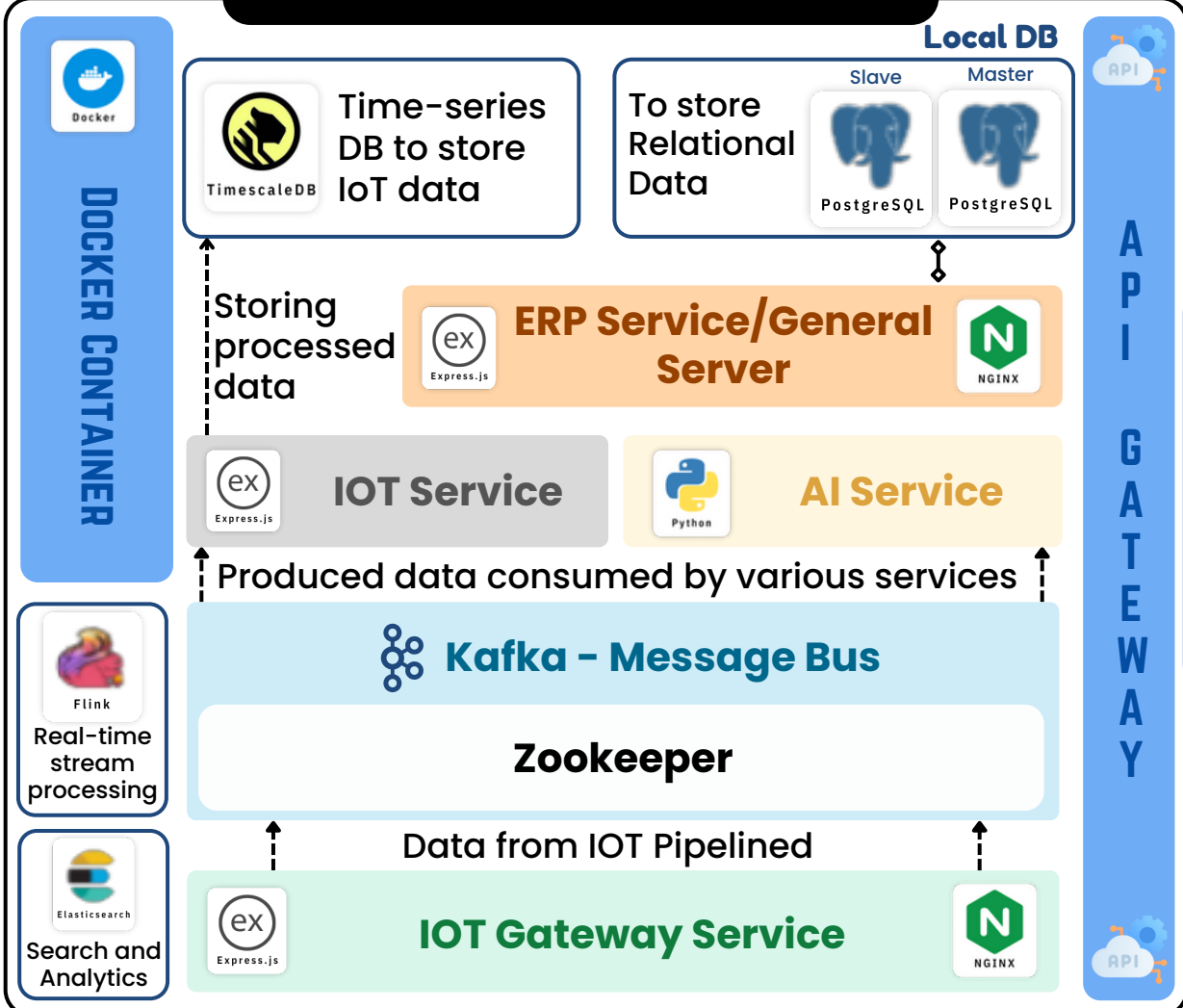
Real-time monitoring of employees

AI - based Shift Handover Recommender

Easy data migration from existing system

## TECHNICAL APPROACH

## 4 BACKEND ARCHITECTURE



## 8 AI COMPONENTS

**AI-powered engine** - Predicts coal production - Forecasts resource needs using real-time mine data.

**Decision Tree algorithm** - Recommends workers for task assignment based on log issues.

F L A S K

3 **ADMIN**

Existing ERP integrated into our system

**Zero Trust Ideology**

Intranet-only access, blocking external connections

**Data Encryption & Hashing**

Data is secured through encryption and hashing

**Google translation API**

**ADMIN**

Efficient Task Handling with Web App for Admin

Easy migration using Data Mapping

Real-Time Critical Alerts for Issue Resolution

Centralized Worker Management

Verification of DGMS Report submission

Converted to digital forms

**OCR**

Automated Text Extraction

Any Type of Log Book

The Safety Management ensures compliance with safety regulations.

**SMP Digitalization**

Log books track daily operations and record important events.

**Manual Logs**

Section-1 ... Section-n

Stakeholders

Supervisor

Operators



1 **COAL MINES**

Coal mines are split into sections consisting of a supervisor and set of workers for smooth flow of operation and safety

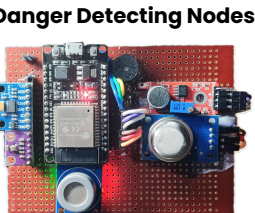
Real-time updates, Shift logs, IoT data via backend

**E2E Encryption**

**Internet** - Data sent directly to the backend

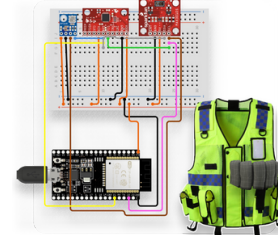
**Fallback**

**No internet** - Data routed via deployed nodes



Node Deployment in Mines

ESP-WIFI-MESH

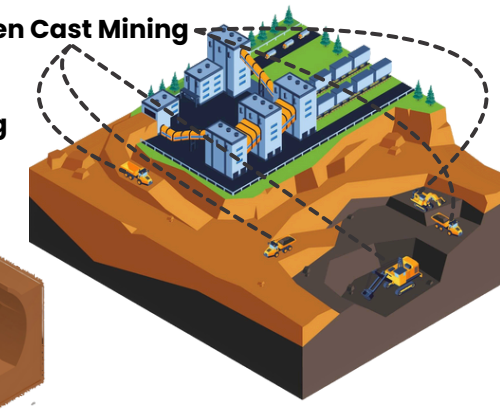


**Safety Vest**

transmit SpO2 and fall detection data to the nearest node for relay.

**Underground Mining**

**Open Cast Mining**



## 5 SHIFT SUPERVISOR

**Web/Desktop App**

Cross-Platform

React.js

Shadcn UI

Typescript

Redux

Mapbox

Socket.io

**Predictive Simulation Engine**

**Real-time IoT logs**

Supervisors can view/Edit Round Plans

Supervisors do Shift Allotment

AI powered assistant to assist Supervisor

Supervisors approve & Submit Shift-Handovers

Streamline Shift Log

DGMS Compliance Log Report Submission with Admin Verification

Automatic PDF report



**DGMS**

Regulatory authority of mining sector in India

Access the Mobile App

## 6 SHIFT OPERATORS

**Mobile App**

Cross-Platform

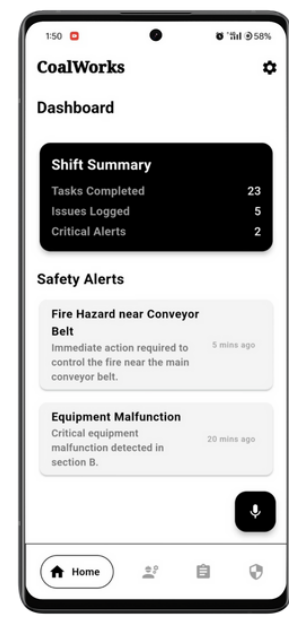
Notification

New task assigned. Click here to view details.

SQL Lite

Bloc

Flutter



Quick summary of shifts, tasks, issues, and alerts

Log details about tasks, issues, alerts during rounds

Support for major Indian languages

Smart voice controls for assistance

# FEASIBILITY AND VIABILITY

## FEASIBILITY



### Technical Feasibility

India's coal industry is expected to **grow by 7.57%** (2024-2029), and our AI/IoT software **improves productivity and safety** in mining.



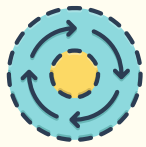
### Operational Feasibility

India's IT spending is at **\$2.48 billion** this year, and our system **digitizes logs and safety**, reducing errors and **increasing efficiency**.



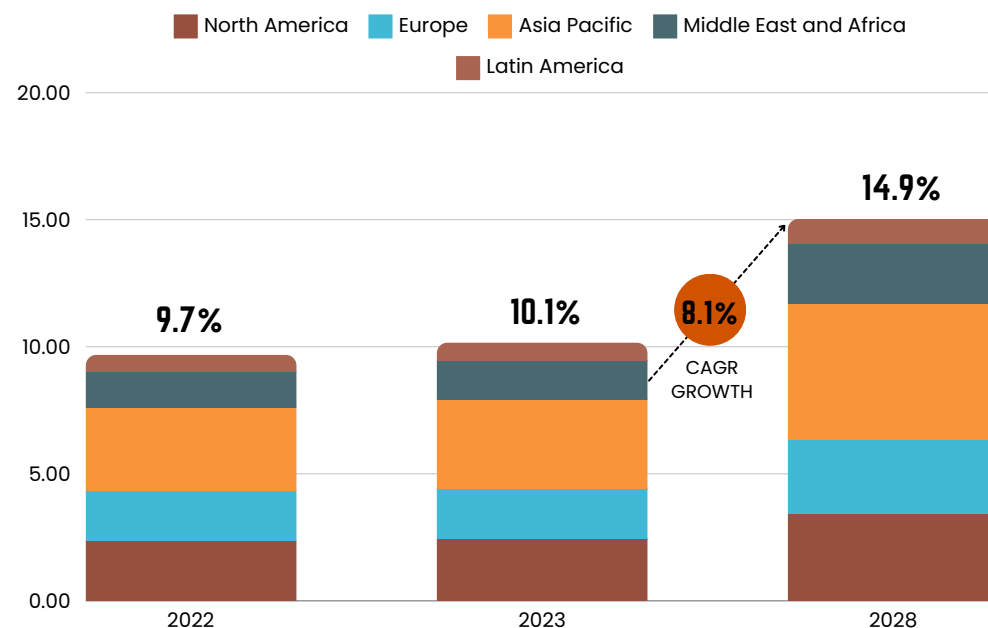
### Economic Feasibility

Our software can be developed in India for **35-50% less** than in the West, making it both **practical and affordable** for the market.



### Regulatory Feasibility

Software **automates DGMS compliance checks**, enhancing safety and easing audits in mining.



### Market Viability

The **Digital India initiative** is driving tech adoption, creating a favorable environment for our software

### Sustainable Viability

The global mining software market is growing at an **8.1% CAGR**, ensuring sustained demand for our product

## CHALLENGES

### Integrating into existing coal mine operations in India and ensuring smooth user adoption



#### Language barriers

- Indian languages
- Regional languages Support



#### System compatibility

- Docker guarantees system consistency.



#### Integration with ERP

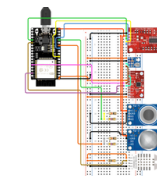
- ERP migration via mapping and APIs



#### Hands-on training

- 2-week training with hands-on practice

### Ensuring actual safety in mines and accurately cross validating workers' productivity



#### Scalable nodes

- Real-time scalable worker safety nodes



#### Vital Sensors

- Sensor vests monitor workers' vital signs



#### AI Analysis

AI engine uses sensor data to track calories burned, distance, and activity metrics

### Managing data communication and ensuring safety and privacy in no-internet zones?



#### Mesh Network

- IoT mesh network for network routing



#### Local DB

- Local databases avoid data loss during outages.



#### Zero Trust Policy

- Access limited to only Verified users



#### E2EE

- End-to-end encryption protects all data

# IMPACTS AND BENEFITS

## Economic Benefits

Though there's an upfront cost, the system pays off by cutting down on downtime and preventing accidents, leading to substantial long-term savings and increased productivity.



## Social Benefits

CoalWorks enhances worker safety and well-being while promoting sustainability through efficient resource management, contributing to safer operations and reduced environmental impact.



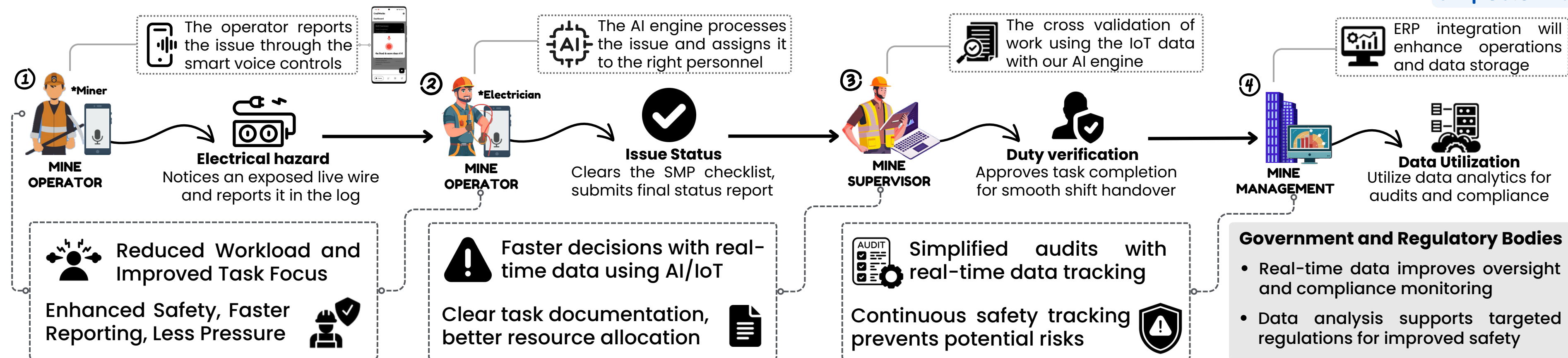
## Environmental Benefits

Improved efficiency and reduced downtime can lead to more effective use of machinery, potentially lowering emissions and reducing the carbon footprint of mining operations.



## STAKEHOLDERS & IMPACTS

### Sample Scenario



## OUR PROMISE

Our solution perfectly aligns with the UN Sustainable Development Goals.



### India's Goal:

Reach 1.5 B Ton by 2030 [Fy April 2030-March 2031]

On 2023 India Produced 997.83 M Tons, so the CAGR is 6.7%

$$\text{CAGR} = (1.504)^{\frac{1}{10}} - 1 \approx 0.067 \text{ or } 6.7\%$$

### \*Consolidated Proof documents of all research is provided in slide 6

CoalWorks is projected to improve **efficiency by 7.69%**, validated against existing solutions.



7.69% increase adds 385.275 million tons, cutting the government's target by 22.67% for 2030.

## COAL MINES IN INDIA - RESEARCH



India's commercial coal mining began in 1774, expanding with steam locomotives in 1853

Coal is a **vital energy source** in India, fueled by steel industry needs and rising energy demands.



## Mining in India

India, second-largest coal producer, 2022

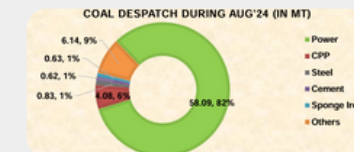
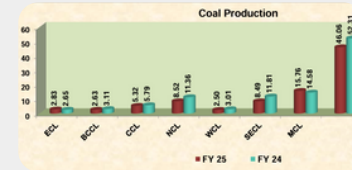
[Click here](#)

## Production Stats

India's position in global coal production

[Click here](#)

India's coal production **dropped 7.51%** to 62.67 MT in Aug'24 from 67.76 MT in Aug'23.



Power utilities despatch **rose 2.92%** to 58.09 MT in Aug'24 from 56.44 MT in Aug'23.

## Coal Mines Regulation and Working

**DGMS** regulates coal mine safety in India, enforcing standards under the Mines Act to protect workers and machinery. It also **conducts inspections, accident investigations, and safety audits to ensure compliance.**

They publishes reports and case studies, which are valuable references for researchers **analyzing mine safety trends, accidents, and technological advancements.**

[Click here](#)

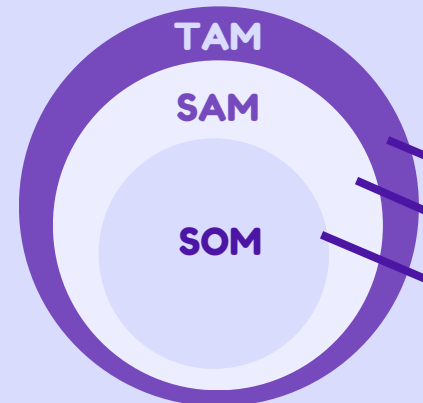
## MARKET - RESEARCH



## Mining Software Market

Analysis of global mining software trends

As of 2023, the global mining software market is projected to reach \$8,334 million by 2030.



TAM - Total Addressable Market  
SAM - Service Available Market  
SOM - Service Obtainable Market

₹ 77,190 crore

₹ 23,157 crore

₹ 1,158 crore

## Mid-Sized Mine Revenue Breakdown

Number of Workers: 500

Safety Vests: 100 units

Mine Danger Detection Nodes: 100 units

## Profit Percentage

Software Sales: Profit margins on software (85%) due to low development cost.

Hardware Sales: Profit margins on hardware (30%) due to material cost.

## Safety Vests

Units: 100

Price per Unit: ₹728

Total: ₹72,800

## Safety Nodes

Units: 100

Price per Unit: ₹720

Total: ₹77,800

## Installation Fees

Units: 200

Price per Unit: ₹980

Total: ₹196,000

## Service and audits

Monthly Cost: ₹5,000

## Total Revenue

Year-1: ₹846,600

[Click here](#)BACK  
END

## Kafka

Kafka's superior scalability sets it apart from other messaging systems

[Click here](#)

## Apache Flink

Flink is a scalable, real-time, distributed stream processing engine

[Click here](#)

## MINE OPERATIONS SOFTWARE - REFERENCES



## Minimax

Tools for optimizing mine planning and scheduling

[Click here](#)

Minimax provides software solutions for mine planning and scheduling, helping mining companies optimize operations and manage costs efficiently.



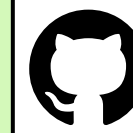
## Innovapptive

Software for Operations, Plant Maintenance, and Warehousing

[Click here](#)

This advanced software is designed to optimize resource allocation and enhance operational efficiency for companies in various sectors.

## Our Works



## GitHub

You can see the Project files of CoalWorks here

[Click here](#)

## Proof Documents

Consolidated proof documents of all the work

[Click here](#)

CLICK ON THE  
IMAGES TO  
VISIT LINKS

## IoT - REFERENCES



## ESP MESH

ESP32: Versatile Wi-Fi and Bluetooth IoT chip

[Click here](#)

## Wifi in coal mines

Wi-Fi enhances safety monitoring in coal mines

[Click here](#)

## AI/ML - REFERENCES



## Intelligent scheduling

Efficiently assigns shifts based on availability

[Click here](#)

## Hazard prediction

Anticipating hazards for preparedness.

[Click here](#)